

SHAWNEE, OK, USA

WMO: 722187

Lat: 35.357N

Lon: 96.943W

Elev: 327

StdP: 97.46

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 93911

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.8	(c) -6.6	(d) -15.1	(e) 1.0	(f) -6.6	(g) -12.6	(h) 1.3	(i) -3.0	(j) 12.1	(k) 7.8	(l) 11.1	(m) 7.7	(n) 3.5	(o) 10	(p) 0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	38.0	24.4	37.0	24.6	35.1	24.6	26.9	33.5	26.2	32.9	25.5	32.2	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.2	30.3	24.2	19.9	28.7	23.7	19.3	28.5	87.1	33.3	82.8	33.3	80.3	32.0	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.9	(b) 9.3	(c) 8.4	(e) -12.9	(f) 39.5	(g) 2.8	(h) 2.8	(i) -14.9	(j) 41.5	(k) -16.6	(l) 43.1	(m) -18.2	(n) 44.7	(o) -20.2	(p) 46.7		
(4) 10.9	(5) 9.3	(6) 8.4	(7) -13.4	(8) 27.5	(9) 2.7	(10) 1.3	(11) -15.3	(12) 28.5	(13) -16.9	(14) 29.2	(15) -18.4	(16) 30.0	(17) -20.4	(18) 30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	4.9	6.8	12.5	16.5	21.2	26.2	28.2	27.9	24.3	17.6	11.8	6.0
	DBStd	9.53	5.99	6.35	5.78	4.92	4.15	2.90	2.96	3.32	3.85	5.14	5.02	5.39
	HDD10.0	537	177	124	39	7	0	0	0	0	0	6	37	147
	HDD18.3	1710	416	324	194	89	22	0	0	0	3	77	201	384
	CDD10.0	3111	20	36	117	204	347	485	563	556	429	241	92	21
	CDD18.3	1241	0	2	13	36	109	235	304	298	183	54	6	1
	CDH23.3	13532	6	17	109	326	947	2512	3681	3556	1838	477	59	3
	CDH26.7	6446	1	2	17	82	318	1157	1970	1918	836	140	6	0
Wind	WSAvg	3.8	3.8	4.0	4.5	4.8	4.1	4.0	3.3	3.2	3.2	3.7	4.0	3.7
Precipitation	PrecAvg	963	35	51	75	98	144	110	57	64	105	105	71	49
	PrecMax	1279	95	160	185	210	365	275	215	215	235	380	208	271
	PrecMin	612	0	0	4	17	30	22	3	5	17	2	1	4
	PrecStd	192	30	38	42	49	84	74	44	48	63	89	53	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	26.0	28.6	32.3	33.7	37.7	42.2	42.6	37.8	32.6	27.5	22.7
		MCWB	12.3	15.9	17.6	19.9	22.4	24.5	24.3	23.9	24.4	22.2	17.7	16.1
	2%	DB	21.0	22.3	26.3	28.6	32.3	36.0	38.6	38.7	35.8	30.3	25.0	19.9
		MCWB	12.6	14.0	16.8	18.6	23.0	24.3	24.1	24.1	24.3	21.5	16.4	14.0
	5%	DB	17.4	19.2	23.9	27.1	30.2	33.9	36.4	37.1	33.0	27.8	22.4	17.1
		MCWB	11.4	13.0	15.9	18.2	22.0	24.6	24.8	24.8	23.6	19.6	15.8	11.9
	10%	DB	14.7	17.1	22.0	25.0	27.9	32.4	34.8	34.9	31.7	26.2	20.4	14.7
		MCWB	8.9	12.1	14.8	17.4	21.0	24.0	24.8	24.7	22.7	19.3	14.9	10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.9	19.7	22.3	25.0	27.4	28.1	28.1	26.4	24.6	20.3	19.0
		MCDB	18.8	21.9	26.0	27.4	31.4	33.4	34.7	35.0	34.3	29.7	23.7	22.1
	2%	WB	14.5	16.2	18.6	21.1	24.0	26.2	26.9	26.9	25.4	23.3	18.9	15.7
		MCDB	18.9	20.1	23.7	25.9	29.2	32.7	33.3	33.7	33.1	27.3	23.3	17.5
	5%	WB	11.6	14.6	17.6	20.1	23.2	25.5	26.1	26.0	24.6	22.3	17.3	12.8
		MCDB	16.2	18.4	21.8	24.6	28.1	31.7	32.5	33.0	31.3	26.0	20.4	15.0
	10%	WB	9.5	12.2	16.3	19.1	22.4	24.8	25.4	25.3	23.8	20.8	15.6	10.9
		MCDB	14.2	16.8	20.5	23.5	27.0	30.7	31.7	32.5	29.5	24.4	19.1	13.5
Mean Daily Temperature Range	5% DB	MDBR	11.7	11.4	12.1	12.1	10.6	10.5	11.2	11.4	11.8	12.1	12.2	10.7
		MCDBR	16.1	15.0	14.5	14.2	12.2	12.0	13.4	13.9	13.3	13.7	14.7	14.5
		MCWBR	9.9	9.3	7.8	7.5	5.6	4.5	4.3	4.4	5.3	6.8	8.4	9.3
	5% WB	MCDBR	14.7	13.4	11.3	11.8	10.5	10.1	10.8	11.7	11.7	10.8	12.3	11.4
		MCWBR	10.5	9.6	6.5	7.3	5.6	4.7	4.7	4.7	4.9	6.2	8.7	9.3
Clear-Sky Solar Irradiance	taub	0.297	0.302	0.343	0.378	0.413	0.425	0.429	0.435	0.388	0.346	0.317	0.294	
	taud	2.504	2.509	2.418	2.334	2.303	2.309	2.334	2.264	2.373	2.483	2.505	2.522	
	Ebn at Noon	911	947	928	906	874	860	853	842	868	883	879	893	
	Edh at Noon	82	91	109	125	131	130	126	133	113	93	82	76	
All-Sky Solar Radiation	RadAvg	2.72	3.30	4.32	5.35	5.78	6.63	6.73	6.12	5.06	3.84	2.87	2.30	
	RadStd	0.28	0.50	0.27	0.39	0.60	0.67	0.36	0.38	0.47	0.50	0.37	0.27	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (36 neighbors)		(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A	(r) N/A	(s) N/A	(t) N/A	(u) N/A

Nomenclature: See separate page